

More Gauss's Law — the Three Symmetries

PHYS 230/235 · Wed Sep 23 · Knight 24.4–24.6

Name: _____

Goal. State Gauss's law and understand what it says, then use it to *compute* $\vec{E}$ — not just check it — for the distributions where symmetry makes that possible: a point charge and a shell, an infinite line, an infinite sheet, and a uniformly charged sphere.

1. What Gauss's law says

Flux Φ counts electric field lines crossing a surface: a line leaving counts +1, a line entering counts −1, so a line that just passes through contributes _____ net. For a **closed** surface,

$$\Phi = \oint \vec{E} \cdot d\vec{A} \left(= \oint E \cos \theta dA, \theta \text{ between } \vec{E} \text{ and the outward normal} \right), \quad d\vec{A} \text{ always points } \underline{\hspace{2cm}}.$$

Gauss's law says that count is fixed by exactly one thing:

$$\Phi = \underline{\hspace{2cm}}$$

not the surface's size, not its shape, not how far it is from the charge.

Check. A point charge sits at the center of a Gaussian sphere. If you double the sphere's radius, does Φ change? Does E at the surface change? (Different questions!)

2. Using it: match the surface to the symmetry

Gauss's law holds for *any* closed surface, but it only lets you *solve* for E when you can pull E outside the integral. Choose a Gaussian surface that shares the charge distribution's own symmetry — **spherical**, **cylindrical**, or **planar** — centered or aligned on the distribution, and two things become true everywhere on it:

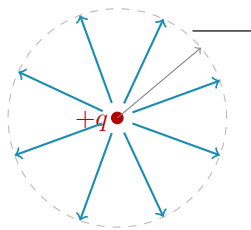
1. E has the same _____ everywhere on it, and
2. $\vec{E}$ is everywhere either _____ or _____ to the surface.

Then $\Phi = E \oint dA = EA$, and Gauss's law becomes plain algebra: $EA = q_{\text{enc}}/\epsilon_0$.

What "symmetric" means: if you closed your eyes while the charge distribution changed in a symmetric way, then opened them again, you couldn't tell anything had changed.

Check. Why does a cube fail as a Gaussian surface for a single point charge?

3. Warm-up: a point charge, then a shell



Label the Gaussian radius r .

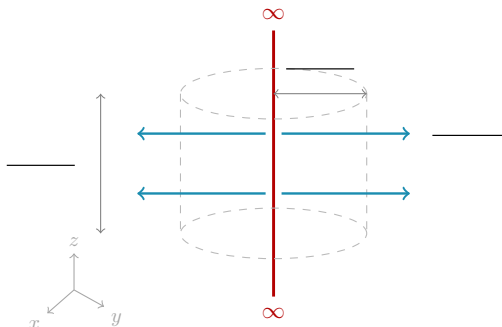
By symmetry $\vec{E}$ is radial and constant in magnitude on the sphere, so $EA = q_{\text{enc}}/\epsilon_0$ with $A = 4\pi r^2$ and $q_{\text{enc}} = q$ gives $E = \underline{\hspace{2cm}}$ — Coulomb's law, recovered from Gauss's law.

Now make it a thin **shell** of charge Q , radius R , instead of a point. **Outside** ($r > R$): the Gaussian sphere encloses $\underline{\hspace{2cm}}$, so $E = \underline{\hspace{2cm}}$ (identical to a point charge — the shell might as well not have a size). **Inside** ($r < R$): the Gaussian sphere encloses $\underline{\hspace{2cm}}$, so $E = \underline{\hspace{2cm}}$.

Check. The sphere you'll work below is *solid*, not a shell — charge fills the whole volume. Will its field inside behave the same way as the shell's?

4. Infinite line of charge

Linear charge density λ . $\vec{E}$ points radially out from the wire and cannot depend on position along it, so the Gaussian surface is a *coaxial cylinder* of radius r and length L .



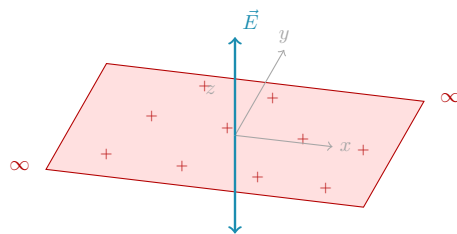
Label the cylinder's radius, its length, and the field arrows. (The wire runs along z — this is a 3D picture, seen at an angle.)

Which part of the cylinder carries flux (and which carries none)? Area A , then q_{enc} , then solve for E . Show L cancelling.

E for an infinite line

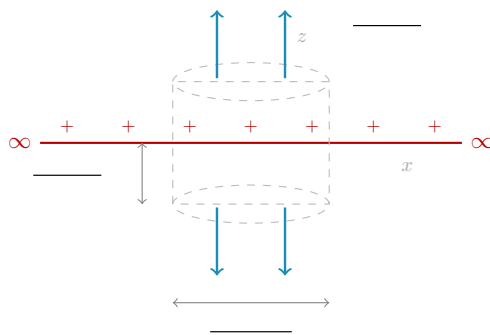
5. Infinite sheet of charge

Surface charge density η . This is a *plane*, not a line: it extends forever in *two* directions, x and y , and $\vec{E}$ points straight away from it along $\pm z$ on both sides.



An infinite sheet, seen at an angle: it extends forever in x and y , and $\vec{E}$ points straight out along z on either side.

Now look at it *edge-on* (sighting along y , so the sheet appears as a line and the pillbox is easy to draw), with end caps of area A_{cap} :



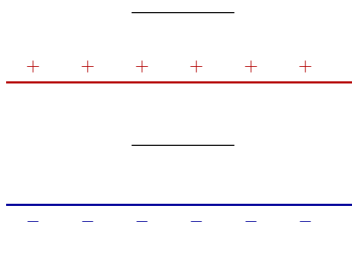
Label the pillbox half-height, the cap area, and the field arrows.

Total area carrying flux, q_{enc} in terms of η and A_{cap} , then E .

E for an infinite sheet

Check. Two surprises. A_{cap} cancelled — why did it *have* to? And the answer contains no r at all. What does that say about walking away from the sheet?

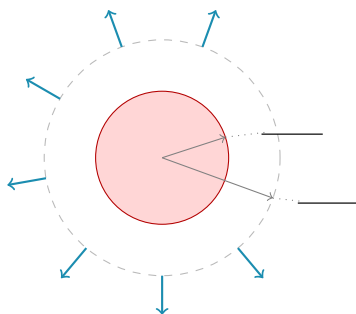
Then: two parallel sheets, one $+\eta$ and one $-\eta$. Superpose the two fields you just found — nothing new to derive.



Write E in each of the three regions: above, between, below.

6. Uniformly charged sphere

Total charge Q spread uniformly through a solid sphere of radius R . There are two regions, and the only difference between them is q_{enc} .



Label the sphere's radius R and the Gaussian radius r .

Outside ($r > R$): the Gaussian sphere encloses _____ of the charge.

q_{enc} , then E . What familiar result does this match?

E outside the sphere

Inside ($r < R$): only the charge within radius r counts. The charge fills the _____, so the fraction enclosed is the ratio of _____.

$q_{\text{enc}} = ?$ then $E = ?$

E inside the sphere

Check. Sketch E against r across both regions. At $r = R$ — jump, kink, or neither?



All three results are in the *Applying Gauss's Law* sim — open its **Info** panel to check yours.